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Article Title: A Comparison of Different Modes of Morning Priming Exercise on Afternoon Performance

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Running title: Comparison of priming exercises

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Abstract

Purpose: To assess the effects of different modes of morning (AM) exercise on afternoon (PM) performance and salivary hormone responses in professional Rugby Union players.

Methods: On four occasions (randomised, crossover design), fifteen professional rugby players provided AM (~08:00h) and PM (~14:00h) saliva samples before PM assessments of countermovement jump height, reaction time and repeated sprint ability. Control (passive rest), Weights (bench press: 5 x 10 repetitions, 75% one repetition-maximum, 90 s intra-set recovery), Cycling (6 x 6 s maximal sprint cycling, 7.5% body mass load, 54 s intra-set recovery) and Running (6 x 40 m maximal sprints, 20 s intra-set recovery) interventions preceded (~5h earlier) PM testing. **Results:** PM sprint performance improved ($P < 0.05$) after Weights ($>0.15 \pm 0.19$ s, $>2.04 \pm 2.46\%$) and Running ($>0.15 \pm 0.17$ s, $>2.12 \pm 2.22\%$), but not Cycling ($P > 0.05$). PM jump height increased following Cycling (0.012 ± 0.009 m, $2.31 \pm 1.76\%$, $P < 0.001$) and Running (0.020 ± 0.009 m, $3.90 \pm 1.79\%$, $P < 0.001$), but not Weights ($P = 0.936$). Reaction time remained unchanged between trials ($P = 0.379$). Relative to Control (131 ± 21 pg·ml⁻¹), PM testosterone was greater in Weights ($+21 \pm 23$ pg·ml⁻¹, $+17 \pm 18\%$, $P = 0.002$) and Running ($+28 \pm 26$ pg·ml⁻¹, $+22 \pm 20\%$, $P = 0.001$), but not Cycling ($P = 0.072$). Salivary cortisol was unaffected by AM exercise ($P = 0.540$). **Conclusions:** All modes of AM exercise improved at least one marker of PM performance but Running appeared the most beneficial to professional Rugby Union players. A rationale therefore exists for preceding PM competition with AM exercise.

Keywords: Ergogenic, potentiation, hormone, rugby

Introduction

Match play within elite team sports commences at varying times throughout the waking day; with kick-offs typically ranging from 11:00 h to 20:00 h. Although optimised sporting performance is subject to a range of intrinsic and extrinsic factors, the influence of circadian rhythmicity is acknowledged (for review see ¹⁻³); with changes in anaerobic physical performance (e.g., force and power expression) occurring at different times of the day^{2, 4}. While kick-off times are likely determined by extraneous factors (including the demands of television;⁵), opportunities exist on the day of competition to influence subsequent performance as athletes may be susceptible to changes in their physical performance as a function of time⁶.

Testosterone and cortisol concentrations exhibit circadian rhythmicity, and are known to correlate with indices of athletic performance ⁷⁻⁹; particularly in elite athletic populations ^{7, 10, 11}. For example, salivary testosterone appears highly correlated with both squat strength ($r = 0.92$) and sprint times ($r = -0.87$) in elite strength trained athletes ¹¹. Moreover, improved three repetition maximum strength was correlated to the acute increase in testosterone concentrations elicited via visual stimulation ⁷ and pre-game testosterone concentrations have been implicated in match outcomes in professional rugby players ¹². However, testosterone and cortisol typically display an early morning (AM) peak before slowly declining across the waking day ^{4, 13}. Considering the potential role of testosterone in mediating athletic performance, offsetting the circadian decline could be of benefit to sporting activities performed at times when testosterone concentrations have experienced a circadian decline, such as in the afternoon (PM). Acutely, a strength/hypertrophy training stimulus can raise post-exercise testosterone concentrations ¹⁴ and thus may be beneficial when preceding subsequent training/competition. However, the effects of other modes of activity (e.g., running and cycling protocols) upon post-exercise testosterone responses remains unclear.

Ekstrand et al.¹⁵ have demonstrated that an AM resistance session that included back squats performed to failure and power clean exercises, improved throwing distance in well trained shot-putters when performed 6 h before subsequent exercise. Similarly, improved PM performance has been observed in Rugby Union players who preceded PM physical performance assessments (i.e., countermovement jumps, 40-m sprints, bench press and back squats) with sprints (5 x 40 m) and whole-body resistance (bench press and back squat routines up to 100% of three repetition-maximum values) exercises 6 h earlier⁶. Notably, the AM sprint and resistance exercise attenuated a circadian decline in testosterone concentrations when compared to a rested control trial⁶. Such findings highlight a potential role for specific modes of AM exercise to improve PM performance, and that such findings may be modulated by changes in hormone status.

Unfortunately, acknowledging the practical considerations associated with the pre-competition practices of professional athletes, the methods of AM exercise examined previously^{6, 15} may preclude their use on the day of competition and/or have limited transfer to match-specific performance indicators. Whole body resistance exercises performed to maximal intensity and/or failure, while beneficial to linear sprinting and force expression, are unlikely to be routinely adopted in the pre-competition setting. Therefore, methods of “priming” PM performances which may be better accepted by players and coaches on the day of competition while demonstrating transfer to match-specific performance indicators are appealing. Moreover, a comprehensive comparison of different priming methods remains to be investigated. Therefore, in a group of professional Rugby Union players, the aim of this study was to compare the effects of three different AM training interventions on PM performance, and profile the associated changes in salivary hormone concentrations. We hypothesised that AM exercise would influence PM performance and salivary hormone responses.

Methods

Subjects

This study received ethical approval from the Swansea University Ethics Committee and was conducted in the spirit of the Helsinki Declaration. Fifteen male Rugby Union players (all competing in back positional roles) recruited from a French top-tier club (age: 24 ± 3 years, mass: 98.2 ± 8.3 kg, stature: 1.86 ± 0.08 m) provided written informed consent before participation.

Design

Players were required to attend the testing venue on six occasions throughout a 28 day period; the first two of these sessions were preliminary visits that included procedural habituation whereas each main trial required a single visit. Four main trials (Control, Weights, Cycling, Running), separated by 5-7 days, were completed in a randomised, counter-balanced and cross-over design. To minimise the effects of circadian variation, the timing of measurements were consistent between trials. A rest day, abstention from caffeine and replication of dietary intake was required in the 24 h before each trial and testing took place in the middle of the competitive season where players were in a maintenance phase of training and preparation.

Main trial procedures

Upon arrival, resting saliva samples were taken (~08:00 h) before players completed the experimental interventions and proceeded to rest for ~5 h thereafter. Four interventions were required throughout the study. For Control, players were instructed to relax (i.e., watch television, read, listen to music) in a temperate environment whereas for Weights, players completed an upper body bench press protocol requiring five sets of 10 repetitions loaded to 75% of their predicted one repetition maximum. Consecutive sets were separated by a 90-s rest interval. For Cycling, six sets of 6-s cycle ergometer sprints (Wattbike Pro, Wattbike Ltd,

Nottingham, UK) that were separated by 54-s of passive rest, were performed against a load of 7.5% of body mass. For Running, six 40 m sprints that required a 180° change of direction at the mid-way point of each sprint was required¹⁶. Each sprint was interspersed with 20-s recovery, until 6 sprints were completed in total. All interventions were preceded by a standardised warm-up (~5 min) that required completion of dynamic stretching, multidirectional and linear speed channel drills, and acceleration and deceleration attempts.

At ~14:00 h, all participants attended the laboratory for assessment of PM performance and physiological responses. In the time between AM and PM testing, players attended a game preview presentation. Collection of an additional saliva sample preceded a reaction time test, three countermovement jump (CMJ) attempts and an assessment of repeated sprint ability. The reaction time test, which consisted of 10 attempts, required players to press a key on a laptop when prompted by a random (i.e., delayed between 2000 – 4000 ms to prevent anticipation) command. Reaction times represent an average for each experimental condition. For CMJ testing, and following a short standardised warm-up (~5 min; including dynamic mobility drills and channel running exercises at various percentages of maximal speed), players performed three maximal CMJ attempts (separated by 30 s) on a portable force platform (Type 92866AA, Kistler, Germany) while keeping arms akimbo. Thereafter, repeated sprint ability testing consisted of six 40 m (two 20 m runs with a 180° change of direction) timed sprints (Brower Timing Systems, Salt Lake City, USA), interspersed with 20-s rest. Total and average sprint times were calculated as per Rampinini et al.¹⁶. Verbal encouragement was provided throughout each physical test.

Jump height was determined using equation 1. The instantaneous velocity and displacement of the player's centre of gravity was derived from the vertical component of the ground reaction force and the participants' body mass.

$$\text{Eq'n 1: Countermovement Jump Height (m)} = (\text{Take off velocity}^2) / (2 \times \text{gravity})$$

Saliva samples were collected into sterile vials (LabServe, New Zealand) via passive drool which were then stored at -80°C . Players were instructed to avoid eating, drinking warm fluids, and brushing of teeth in the two hours preceding sampling in order to minimise sample dilution. Samples were analysed in duplicate using commercially available enzyme immunoassay kits (Salimetrics LLC, State College, PA, USA). The lowest detection limits for testosterone and cortisol were $0.001\text{ nmol}\cdot\text{L}^{-1}$ and $0.08\text{ nmol}\cdot\text{L}^{-1}$, respectively and inter-assay CV values were $<10\%$ in both cases.

Statistical analyses

Statistical analyses were carried out using SPSS Statistics software (IBM Inc., USA) with significance set at $P\leq 0.05$. Data are reported as mean $\pm$ standard deviation (SD). For single time-point data, one-way analyses of variance (ANOVA) were performed whereas for data expressed over multiple time-points, two-way repeated measures ANOVAs (within-participant factors: trial $\times$ time) were performed. Where significant interaction effects were observed, trial was deemed to have influenced responses and simple main effect analyses were undertaken. Partial eta-squared (η^2) values were calculated and LSD *post-hoc* tests were performed to isolate significant differences.

Results

Repeated sprint performance

Figure 1 presents the repeated sprint data as a function of trial. Although mean ($7.56 \pm 0.28\text{ s}$, $F_{(2,27)} = 1.749$, $P=0.192$, $\eta^2 = 0.119$) and total ($45.33 \pm 1.69\text{ s}$, $F_{(2,27)} = 1.766$, $P=0.189$, $\eta^2 = 0.120$) repeated sprint times were comparable to Control, individual sprints were influenced by the mode of AM exercise (time $\times$ trial interaction: $F_{(6,81)} = 3.578$, $P=0.003$, $\eta^2 = 0.216$). When compared to Control, the first two sprints were faster in Weights (sprint 1: $0.16 \pm 0.17\text{ s}$, $2.28 \pm 2.25\%$, $P=0.003$; sprint 2: $0.15 \pm 0.19\text{ s}$, $2.04 \pm 2.46\%$, $P=0.010$) and Running (sprint 1: $0.15 \pm 0.17\text{ s}$, $2.12 \pm 2.22\%$, $P=0.005$; sprint 2: $0.17 \pm 0.20\text{ s}$, $2.36 \pm$

2.57%, $P=0.006$) but not Cycling ($P=0.171$, $P=0.884$, respectively). No differences existed between trials from sprint 3 onwards (all $P>0.05$) whereas timing effects were observed after sprint 1 ($F_{(2,21)} = 119.062$, $P<0.001$, $\eta^2 = 0.902$).

Countermovement jump height and reaction time performance

Jump height was influenced by trial ($F_{(3,42)} = 18.806$, $P<0.001$, $\eta^2 = 0.573$). Relative to Control (0.521 ± 0.041 m), increased jump heights were observed in Cycling (0.012 ± 0.009 m, $2.31 \pm 1.76\%$, $P<0.001$) and Running (0.020 ± 0.009 m, $3.90 \pm 1.79\%$, $P<0.001$), but not Weights ($P=0.936$). AM exercise did not affect reaction time ($F_{(2,23)} = 0.970$, $P=0.379$, $\eta^2 = 0.065$) when compared to Control (266 ± 34 ms).

Salivary hormones

Testosterone concentrations were influenced by trial (trial x treatment interaction: $F_{(2,23)} = 11.177$, $P=0.001$, partial- $\eta^2 = 0.444$) and time of sample (time effect: $F_{(1,14)} = 252.164$, $P<0.001$, $\eta^2 = 0.947$). While AM testosterone concentrations were similar between trials ($P=0.176$), PM values were greater in Weights ($+21 \pm 23$ pg·ml⁻¹, $+17 \pm 18\%$, $P=0.002$) and Running ($+28 \pm 26$ pg·ml⁻¹, $+22 \pm 20\%$, $P=0.001$), but not Cycling ($P=0.072$), when compared to Control (131 ± 21 pg·ml⁻¹). Notably, all interventions attenuated the reduction in testosterone from AM to PM observed in Control (-53 ± 23 pg·ml⁻¹) but Running best preserved PM testosterone when compared to all other trials (Figure 2).

Salivary cortisol concentrations did not differ according to trial (time x trial interaction: $F_{(2,28)} = 0.626$, $P=0.540$, $\eta^2 = 0.043$) but did vary due to sampling time (time effect: $F_{(1,14)} = 307.048$, $P<0.001$, $\eta^2 = 0.956$) with concentrations reducing ($P<0.001$) from AM (0.412 ± 0.091 ng·ml⁻¹) to PM (0.332 ± 0.090 ng·ml⁻¹).

Discussion

In professional Rugby Union players, this study compared the efficacy of different modes of AM exercise on indices of PM performance, and profiled the associated salivary hormone responses. In agreement with our hypothesis, AM exercise influenced PM performance and salivary hormone responses. Moreover, our data also highlights that six 40 m sprints (incorporating 180° changes of direction) performed ~5 h before subsequent exercise improved both sprinting and jumping performance while eliciting the greatest effect on salivary testosterone concentrations. Additionally, upper-body resistance exercise augmented sprint performance and elicited more favourable testosterone concentrations whereas cycle sprints increased countermovement jump performance alone. Our findings extend the concept that the morning of competition provides an additional opportunity in the competitive cycle whereby subsequent performance can be enhanced.

Improved strength and power performances and an attenuated circadian decline in testosterone concentrations have been reported for 6 h after heavy (i.e., up to 100% of three repetition maximum) bench press and back squat exercises ⁶. Similarly, shot-putters who performed an AM resistance training session that included back squats to failure improved their PM throwing distances achieved 6 h later. Although ergogenic, these interventions may not be well accepted in the pre-competition setting. Alternatively, we provide evidence that an upper body resistance session aiming to elicit a hypertrophy response, a repeated sprint running protocol requiring changes of direction, and an off-feet sprint cycle session appear beneficial to either indices of performance integral to team sports, and/or salivary hormone responses that have been positively associated with physical performances ^{7, 11}. However, as no intervention examined here elicited any differences in performance of the reaction time task, our findings do not support the notion of a positive role of testosterone on cognitive performance that has been reported previously ¹⁰.

Repeated sprint running attenuated the declines in PM testosterone concentrations observed in Control; a finding which is contrary to those of Cook et al.⁶ who observed no effect of five sets of 40 m linear sprints on PM salivary testosterone. Notably, when compared to linear movement, a greater intensity of submaximal shuttle running is reported when changes of direction are required¹⁷; a response which demonstrates a dose-response with the number of directional changes needed. It is therefore plausible that the intensity of the Running trial in this study was greater than that required by Cook et al.⁶ and thus contributes somewhat to explaining the differences between studies in the testosterone responses observed. Intensity, albeit in resistance training, has been linked to post-exercise testosterone concentrations¹⁸.

We report beneficial effects of repeated sprint running on high intensity actions (e.g., sprinting and jumping) performed 5 h later. In contrast to previous authors⁶, we observed improved jump height as well as enhanced sprint speed after the Running trial; findings which indicate that the benefits of prior exercise, specifically repeated sprint running, may not be completely task-specific. Although the explanation for these differences between the studies remains unclear, the lowest reduction in testosterone concentrations from AM to PM occurred after the Running intervention.

In the case of sprint running performance, where observed, improvements were isolated to the first two sprints of the six-sprint protocol. In comparison to studies that have examined the effects of AM exercise on isolated bouts of PM activity^{6, 15}, this is the first study to report the effects of AM exercise using repeated high intensity efforts separated by short (i.e., < 30 s) periods of recovery. It is likely that the demands of the first two sprints were of sufficient intensity to offset the benefits associated with “primed” exercise performance. Nevertheless, while highlighting that performance at the start of a match is most likely improved by the Running and Weights interventions used here, it is worth noting

that the influence of AM exercise on performance throughout the full duration of a match is unclear and therefore provides future research opportunities. An initial starting point for researchers interested in this area may be to examine the effects of AM interventions on PM performance using simulated match-play protocols.

Data from the Running and Weights trials support observations that acute levels of free testosterone appear predictive of subsequent physical performance^{8, 11, 12}. However, the absence of statistically significant changes in testosterone concentrations in Cycling, despite improved PM jump performance, suggest that alternative mechanisms may explain the efficacy of improved PM performance following AM exercise. Although testosterone has been reported to mediate behavioural motivation¹⁹, core and muscle temperature are also known to vary throughout the day^{1, 20} and to acutely modulate muscular performance²¹. In the absence of body temperature measurements, one cannot discount the possibility that this confounding variable elicited somewhat of an effect. Indeed, many other physiological processes (e.g. oxygen consumption, ventilation, heart rate) important to human performance are also known to vary with the time of day²².

Practical Implications

In professional Rugby Union players, our data highlights that AM exercise offers an ergogenic strategy for improved PM performance when a period of ~5 h separates exercise bouts. Specifically, six 40 m sprints (including a mid-sprint 180° change of direction) increased subsequent sprint and jump performance and attenuated circadian declines in testosterone concentrations thereafter. Upper body resistance exercise and/or cycle sprinting elicited more favourable salivary hormone profiles when compared to rest and thus offer an alternative to repeated sprinting if not logistically feasible.

Conclusions

We report novel data concerning the efficacy of different modes of AM exercise on PM performance and salivary hormone responses. When performed ~5 h before subsequent exercise, six 40 m sprints (requiring a 180° mid-way change of direction and separated by 20 s recovery) elicited the greatest attenuation of PM testosterone concentrations and improved lower body explosive actions (i.e., countermovement jump and sprinting performances) performed thereafter. If not logistically possible, our results also highlight that applied practitioners may wish to consider the use of upper body resistance exercise (i.e., five sets of 10 repetitions at 75% of one repetition maximum, separated by a 90-s rest period) or off-feet cycle sprinting (i.e., six sets of 6-s cycle ergometer sprints against a load of 7.5% of body mass, separated by 54-s of passive rest) on the morning of competition as beneficial effects were observed on PM salivary testosterone concentrations and/or physical performance markers following these interventions.

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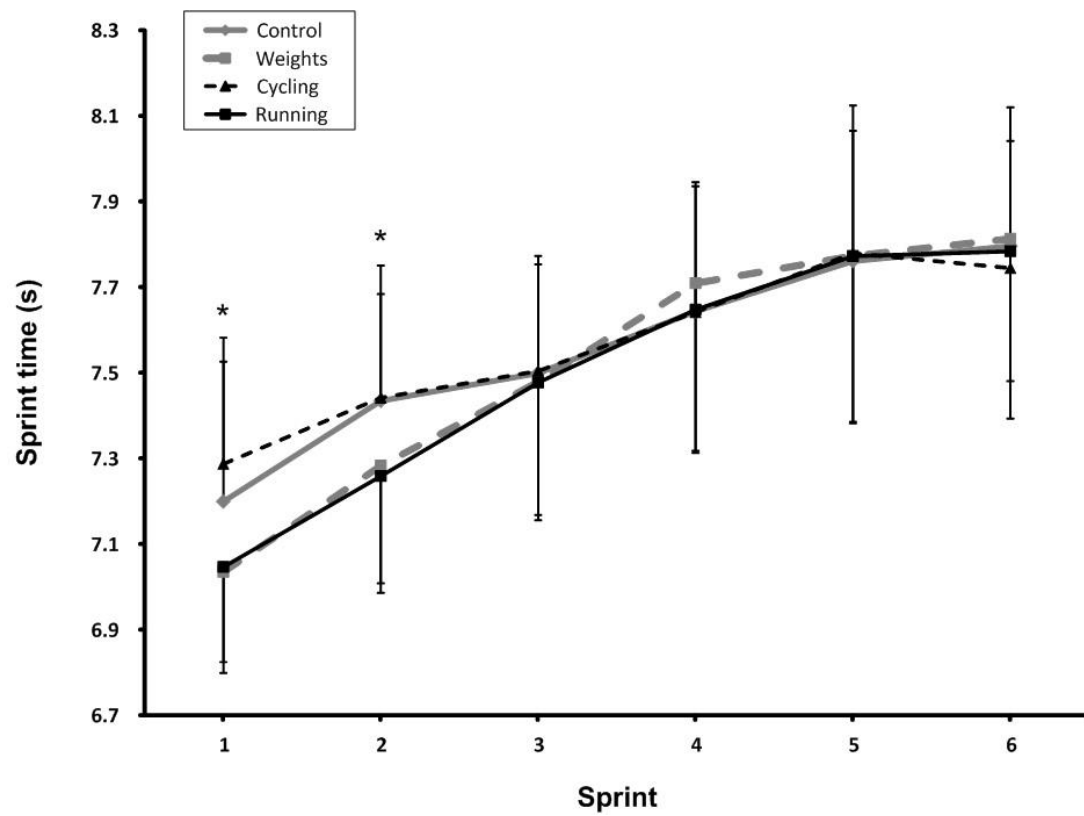


Figure 1: Mean $\pm$ SD sprint times throughout the Control, Weights, Cycling and Running trials. * indicates significant main effect of trial ($P < 0.05$) at the corresponding time-point.

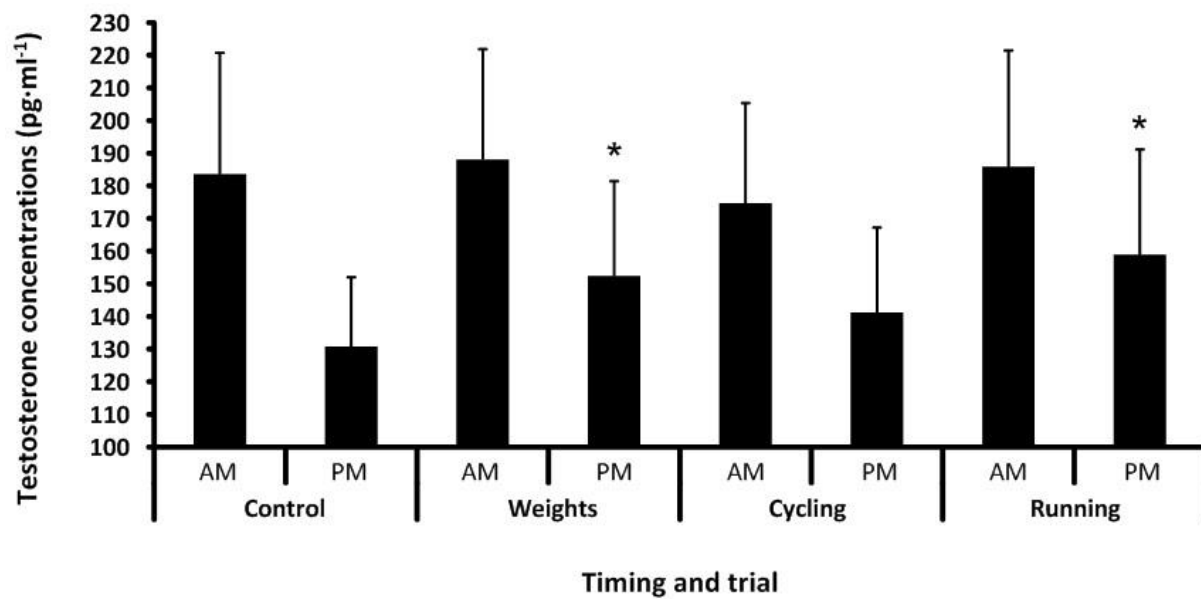


Figure 2: Mean $\pm$ SD testosterone concentrations throughout the Control, Weights, Cycling and Running trials. AM represents morning, PM represents afternoon, * indicates significant difference ($P<0.05$) compared to corresponding time-point in Control.